

VEDHA PRAMEEDH VADDEPALLY

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EDUCATION

The University of Texas at Dallas, Richardson, TX
Bachelor of Science, Computer Science

May 2028

SKILLS

Technical Skills: Python, Java, C++, React.js, React Native, Vite, Express.js, Flask, PyTorch, YOLOv8, scikit-learn, Expo, Git
Interests: AI/ML, Full-Stack Development, Assistive Technology, Local AI Infrastructure, Hardware

EXPERIENCE

Vertically Integrated Projects (VIP) – UTD, Richardson, TX
Undergraduate Research Member

Jan 2025 – Present

- Built a **multi-task deep learning pipeline** combining vision transformers and BERT-based NLP to automate structured report generation from medical imaging data.
- Collaborating with clinical partners to develop and validate AI pipelines on a large medical imaging dataset under a formal data use agreement.

PROJECTS

77 GHz Automotive Radar – Short-Range Obstacle Detection | IEEE

Fall 2024

- Designed a **77 GHz pulse-radar system using the TI IWR1443 single-chip mmWave sensor** with patch microstrip antennas to detect frontal obstacles in real time, overcoming optical camera limitations in adverse weather.
- Simulated antenna performance in **Ansys HFSS**; transmitted sensor data over a **CAN Bus** to display a real-time 2D polar proximity graph on a monitor.
- Presented to a panel of faculty professors; validated system accuracy through **real-world and simulated environment testing** assessing range resolution, object trajectory, and detection reliability.

PokéLearn – HackAI 2026

Spring 2026

- Built a **browser-based learning RPG** in React 18 + Vite that gamifies data science education using Pokémon-style mechanics across 10 topic tracks.
- Designed a **four-tier adaptive placement system** (Beginner → Master) with a 12-question pretest; shuffled 200+ questions using Fisher-Yates to eliminate answer-position bias.
- Integrated **Groq API (LLaMA 3.1) delivering ~250 tokens/sec** for near-instant in-app AI tutoring responses.

FutureKey – AI Real Estate Intelligence Platform

Spring 2026

- Built a **React + Django full-stack platform** for the Seattle/King County housing market with interactive maps, property search, and a bid estimation feature.
- Trained an **XGBoost price forecast model** on geocoded housing data; served predictions via a Django REST API to power buyer and investor decision workflows.

TrustEstate AI – HackUTD 2025

Nov 2025

- Placed **2nd out of 369 projects** in the CBRE sponsor challenge at HackUTD (1,204 participants, North America's largest 24-hour hackathon).
- Co-built an **AI real estate assistant with Gemini-backed confidence scoring** across source quality, recency, and data-gap dimensions via a Google Maps interface.

Pediatric X-Ray Cancer Detection – AIMD (2nd Place)

Spring 2025

- Trained a CNN classifier on the **Stanford CheXpert dataset (224,316 chest X-rays, 65,240 patients)** to flag potential cancer indicators across 14 pathology labels.
- Built **end-to-end preprocessing and evaluation pipelines** in PyTorch; placed 2nd in the AIMD Annual ML Challenge.

HONORS AND AWARDS

- **2nd Place, CBRE Sponsor Challenge** – HackUTD 2025 (1,204 participants — North America's largest 24-hour hackathon; 369 projects submitted)
- **2nd Place, AIMD Annual ML Challenge** – Pediatric X-Ray Cancer Detection (2025)